



ZOTAC MAG

ZOTAC enters net-top business with its compact, low-powered solution called the MAG



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NOVEMBER 2009



Galaxy GeForce GTS 250

Mid-range cards up to Rs. 10,000

This was the most populous category by far and with good reason – 80 per cent of the gaming fraternity will look to spend within this range. No fools themselves, ATi and NVIDIA are wise to the fact. Such cards are more than suitable for gaming and you can buy a fair bit of graphics grunt for under Rs. 10,000 these days. A sweet, shiny-new entrant to this category is ATi's shiny new Radeon HD 5770. Codenamed 'Juniper', this card is a DX 11 part and surprised us (spoiler!). The other GPUs falling into this price range are the NVIDIA GeForce GTS 250, 9800 GTX and 9800 GT, and ATi Radeon HD 4850, 4830 and 4770.

Sapphire is careful with their

HD 4850 cards' cooling solution. The infamous single-slot cooler was woefully inadequate and these cards heated a lot. It wasn't uncommon to see core temperatures soar northwards of 90 degrees. Sapphire's custom cooler with aluminium fins and a small shroud only covers the core but it is a dual-slot design and obviously better than the stock solution. XFX's HD 4850 (HD-485X-YD HD) has a heatsink and shroud like NVIDIA's cards – it's large, runs along the entire card and is dual-slot design – in fact, XFX seems to be using this very same design on a lot of their cards. ZOTAC's GTS 250 512 MB is a small card – no bigger than the PCB of a 9600 GT and it has a neat custom cooler with large aluminium fins. Unfortunately, it also gets

pretty hot, the body of the cooler and the rear of the card also heats up. The ZOTAC 9800 GT Synergy has no requirement for additional power and makes do with the power supplied by the PCIe slot. The cooler on this card has a large surface area and covers nearly the whole card with a lot of fins and a small fan. Palit's GTS 250 has two 6-pin molex power connects unlike the two ZOTAC cards that have a single 8-pin connect. ASUS has designed a neat cooler for their MATRIX series and they persist with the same design for all cards which is a bit disorienting since there is no NVIDIA or ATi badging to tell, say a Radeon HD 4850 apart from a GeForce 9800GT.

Galaxy's GTS 250 cards have a unique cooler – neat design and slightly smaller than a full length cooler as the shroud is a lot smaller but the cards get slightly hotter than stock-cooler bearing GTS 250 cards.

MSI 9800 GTX (N9800GTX-T2D512) has an issue with the cooler shroud foiling with the retention on the 6-pin molex power connects although the flexible shroud bends slightly

HOW WE TESTED

Our test configuration had to be the fastest around since we needed to minimise bottlenecks. So we chose premium components throughout. Our unanimous choice for the CPU was Intel's Core i7 975 Extreme (3.33 GHz). For the motherboard, we used our last month's Best Performer, the ASUS Rampage 2 Extreme motherboard (Intel X58 chipset-based). This is a uber feature-rich board that's packed to the gills with quality components. Kingston's HyperX tri-channel DDR3 kit (3 x 1 GB, 1600 MHz @ 9-9-9-24) and a 10,000 rpm, 300 GB Western Digital Velociraptor hard drive were also used. Corsair's excellent HX620 provided the juice to power this system. For the display component, we used HP's 2475 LCD monitor, a 24-inch display based on an H-IPS panel.

In the BIOS, we disabled all dynamic stepping, multiplier and voltage settings thereby ensuring the i7 975E processor ran unimpeded at 3.33 GHz all the time. The memory was set at 1600 MHz at timings of 9-9-9-24. Two partitions were created. We used a 32-bit version of Vista Ultimate and disabled all unnecessary features like System Restore and Remote Logging. Unnecessary services were also disabled and a 6 GB permanent swap file was created on C: drive.

All games and benchmarks were installed on D: drive and after install we patched everything to their respective latest versions. Once installation was done, a disk cleanup was run on both partitions. Both partitions were then checked for errors and defragmented.

For game benchmarks, we wanted something that was fairly new with really good engines. The challenge was selecting titles that are also popular. We tried a lot of games, around 15 in all and tested each game on four cards, two ATi and two NVIDIA. This allowed us to check scaling on cards from both vendors. We eventually selected six of the best games on the basis of how well they scaled in performance, the visual appeal of the engine and of course the popularity of the game. We also watched out if any game seemed to favour a particular vendor. Such games were removed to prevent giving an unfair advantage to either ATi or NVIDIA. Latest versions of all games were used. 3D Mark Vantage Professional was patched to its latest version 1.0.2.1.

We segregated cards into two categories on the basis of what chipsets they were based on and not primarily on price (pricing was the secondary factor). Similar GPUs from AMD and NVIDIA were compared –

this means the Radeon HD 4850 was compared to the GeForce 9800 GT and the GTX, while the Radeon HD 5870 was compared to the Radeon 4870 x2 and the GTX 285 / 295 combo. This ensured that a costly 9800 GTX (for example) would find itself with other 9800 GTX cards and not compared to a GTX 260. Within a single category, a division on the basis of price was made.

For the high-end cards above the Rs. 10,000 price mark, we used higher resolutions – 1920 x 1200 and 1680 x 1050 – common for 24 and 22-inch LCD monitors. This screen size is affordable for someone spending on a high-end card. For lower graphics cards that were in the price range of Rs. 7,001 to Rs. 10,000, we used resolutions of 1680 x 1050 pixels and 1280 x 1024 pixels, but at the same visual settings for the sake of comparison. Similarly, FSAA was used at 4x for all cards. Shader detail and other effects were kept at maximum. For benchmarks that allowed it, we recorded the maximum and minimum fps as well as the de facto average fps. Although the maximum value is pretty useless for anything more than a cursory comparison with pretty figures, the minimum fps gives a realistic look at how fast the game will run on that

card. In general anything above 30 fps is playable, although for first person action games we recommend a frame rate in the range of 50 to 60.

We used Riva Tuner's latest version to record temperatures and also checked the drivers reading for the same to check for discrepancies. The cards specifications were noted by using the latest version of GPU-Z. We used the latest version of drivers from NVIDIA's and ATi's websites. For NVIDIA, this was version 190.62 while for the Radeon cards, we used the newest driver for the HD 58xx series since it works fine with the older 4xxx series of cards as well.

Besides raw fps values, we also used took two screenshots each from *Crysis Warhead* and *S.T.A.L.K.E.R. Clear Sky* for visual comparisons. Taken at fixed points in the timedemo benchmark, we used a resolution of 1920 x 1200 pixels, with maximum Anisotropic Filtering and 4x Antialiasing and all effects set to maximum. The score for each game marked on a scale of 10 is a result of aggregates of visual quality – Specular Highlights and HDR lighting quality, Shader Detail, shadow detail and quality, MSAA and its effectiveness as well as particle effects like smoke, fog and explosion effects, quality of body physics.